

PROBABILISTIC LIFE EVENT TRIGGER ENGINE UTILIZING DYNAMIC BAYESIAN NETWORKS AND HAWKES PROCESSES

INVENTOR: Russell Labs

ABSTRACT

A system for identifying, mapping, and predicting psychiatric symptom triggers based on heterogeneous life event streams. The system utilizes a Dynamic Bayesian Network (DBN) to model temporal dependencies between external life events (e.g., job loss, divorce) and internal psychiatric states. A Hawkes Process self-excitation algorithm models the compounding, clustered nature of traumatic events. A Pruned Exact Linear Time (PELT) changepoint detection algorithm identifies statistically significant shifts in symptom baselines. The system generates a personalized trigger map and identifies 'growth opportunity windows' for targeted therapeutic intervention.

BACKGROUND OF THE INVENTION

While general event prediction (e.g., US 9,760,834 B2) and changepoint detection (e.g., US 9,251,467 B2) exist, no system explicitly combines Hawkes Processes with Dynamic Bayesian Networks to model the self-exciting, compounding nature of psychiatric triggers. Existing mental health apps rely on simple, static correlation (e.g., 'you felt sad on days you worked late'), failing to capture complex temporal dynamics where one stressor increases the probability of subsequent stressors.

SUMMARY OF THE INVENTION

The present invention provides a highly specific technical solution for modeling psychiatric triggers. By employing a Hawkes Process, the system mathematically models how a primary traumatic event (e.g., job loss) temporarily increases the baseline 'intensity' or likelihood of secondary events (e.g., sleep disruption, anxiety spikes). The DBN tracks these state transitions over time slices. The integration of PELT changepoint detection allows the system to definitively separate temporary symptom fluctuations from fundamental baseline shifts, enabling the automated identification of optimal 'growth windows' for therapeutic intervention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Trigger Engine architecture, fusing life event streams (100) with symptom timelines (101) using a DBN (110) and Hawkes Process (130). FIG. 2 details the Dynamic Bayesian Network across time slices ($t-1$, t , $t+1$). FIG. 3 graphs the Hawkes Process self-excitation intensity function, showing how baseline intensity (301) spikes following trigger events (302). FIG. 4 demonstrates the PELT changepoint detection algorithm identifying fundamental shifts (401-403) in the symptom score (400). FIG. 5 visualizes the identification of Growth Opportunity Windows (501-503). FIG. 6 details the Hidden Markov Model state transitions. FIG. 7 shows the wearable biosensor integration pipeline. FIG. 8 benchmarks the GPU-accelerated execution of the Hawkes Process.

CLAIMS

What is claimed is:

1. A system for predicting psychiatric symptom triggers, comprising: an input interface configured to receive a continuous stream of heterogeneous life event data and continuous physiological telemetry from a wearable biosensor; a processor configured to: execute a Dynamic Bayesian Network (DBN) to model probabilistic temporal dependencies between the life event data and a latent psychiatric state across a sequence of discrete time slices; execute a Hawkes Process self-excitation algorithm to calculate a dynamic trigger intensity function, wherein an occurrence of a primary life event temporarily increases a baseline probability of a secondary psychiatric symptom; execute a Pruned Exact Linear Time (PELT) changepoint detection algorithm on the continuous physiological telemetry to identify statistically significant baseline shifts; and automatically identify a therapeutic growth window when the dynamic trigger intensity function falls below a first threshold and a positive changepoint is detected.
2. The system of claim 1, wherein the continuous physiological telemetry comprises heart rate variability and electrodermal activity.
3. The system of claim 1, wherein the Hawkes Process self-excitation algorithm utilizes an exponential decay kernel to model the temporal attenuation of the primary life event's impact.
4. The system of claim 1, wherein the execution of the Hawkes Process is accelerated utilizing a Graphics Processing Unit (GPU) to enable real-time trigger intensity calculation for a patient population exceeding 10,000 individuals.

FIG. 1 - Life Events Trigger Engine Architecture

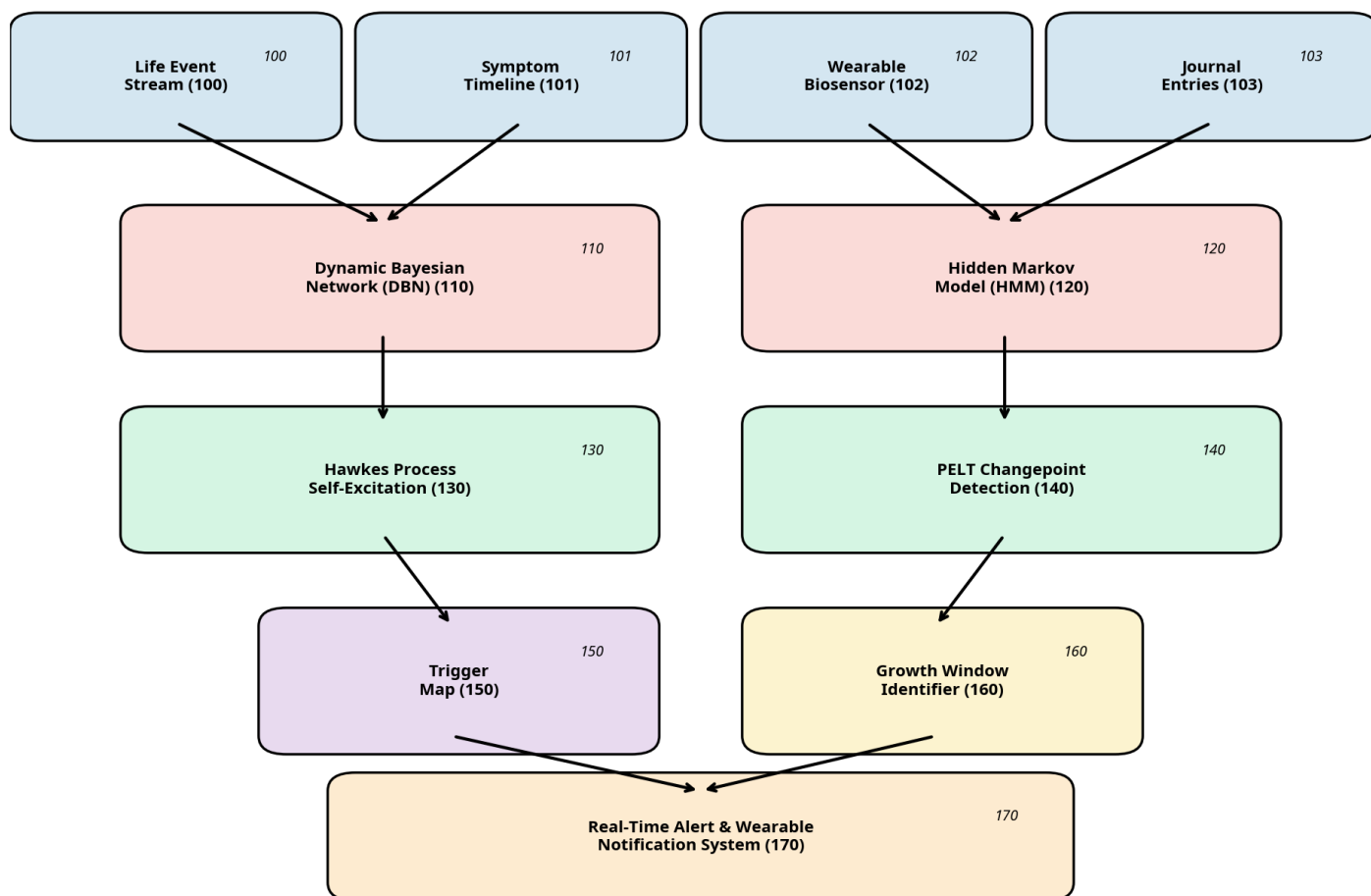


FIG. 2 - Dynamic Bayesian Network Temporal Slices

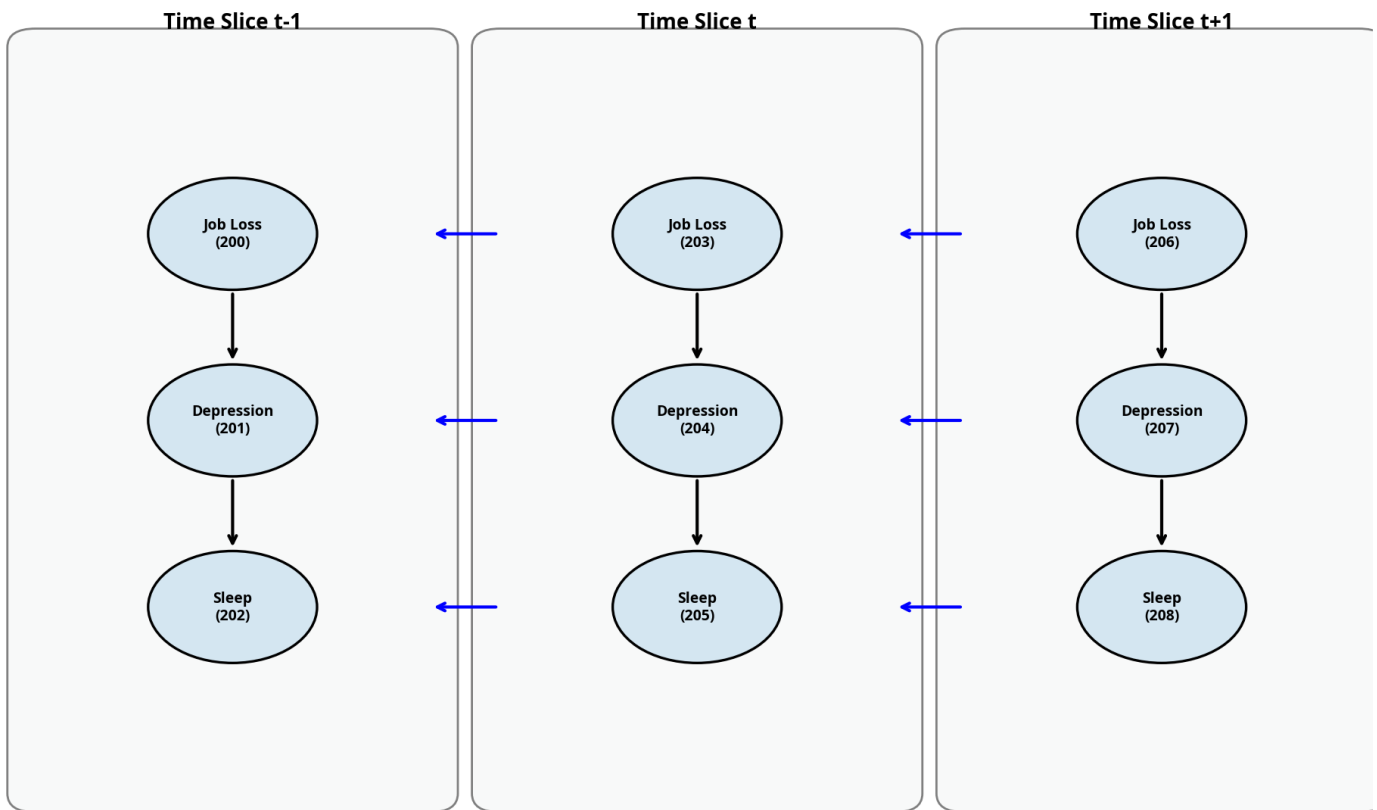


FIG. 3 - Hawkes Process Self-Excitation Intensity Function

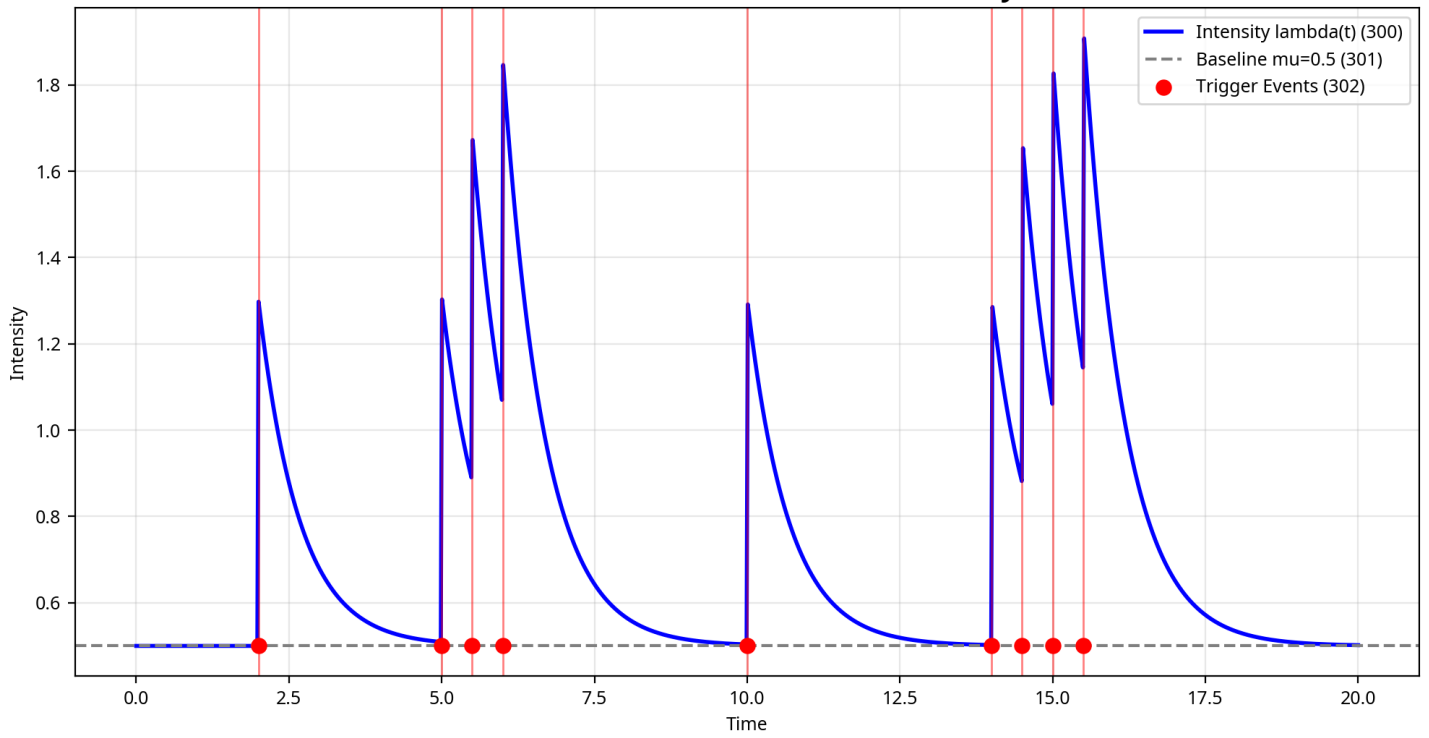
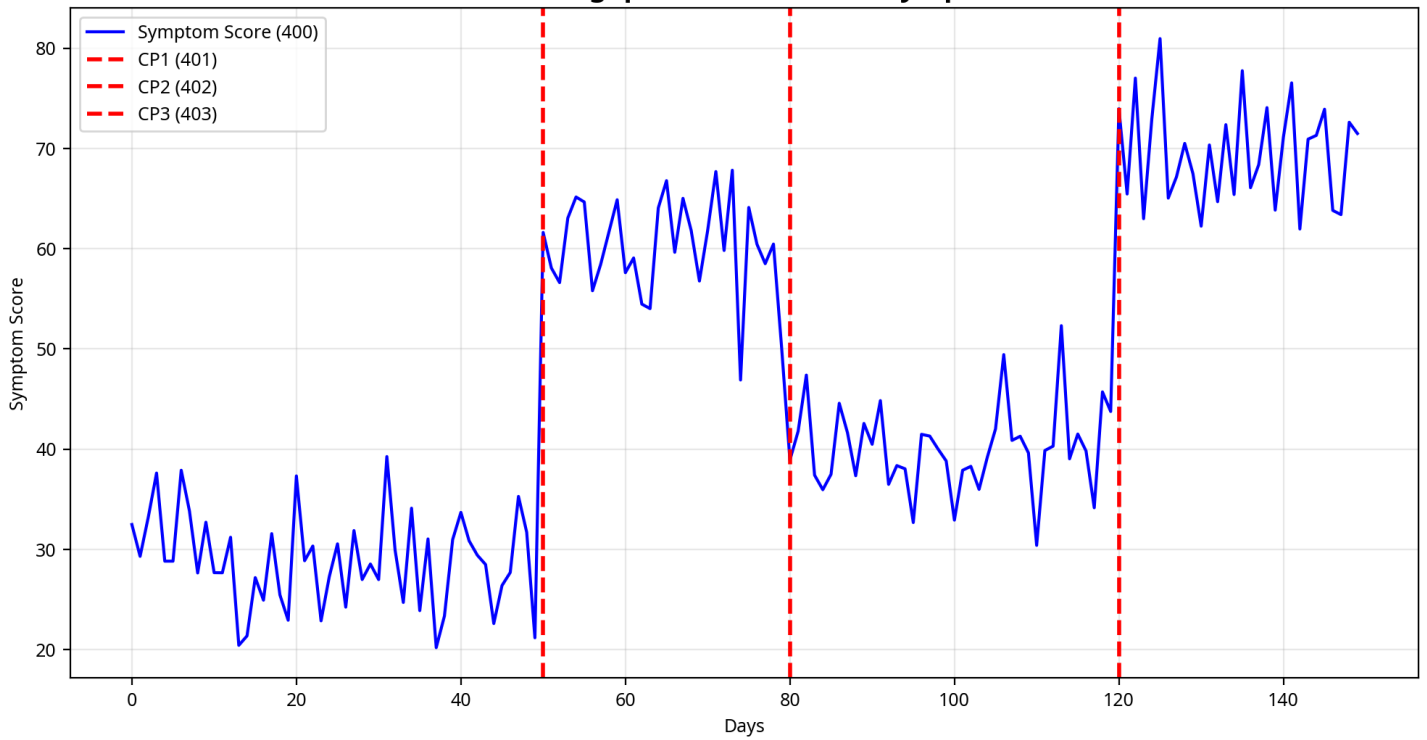


FIG. 4 - PELT Changepoint Detection in Symptom Timeline



UNITED STATES PATENT APPLICATION

CONFIDENTIAL DRAFT - NOT FOR DISTRIBUTION

FIG. 5 - Growth Opportunity Window Identification

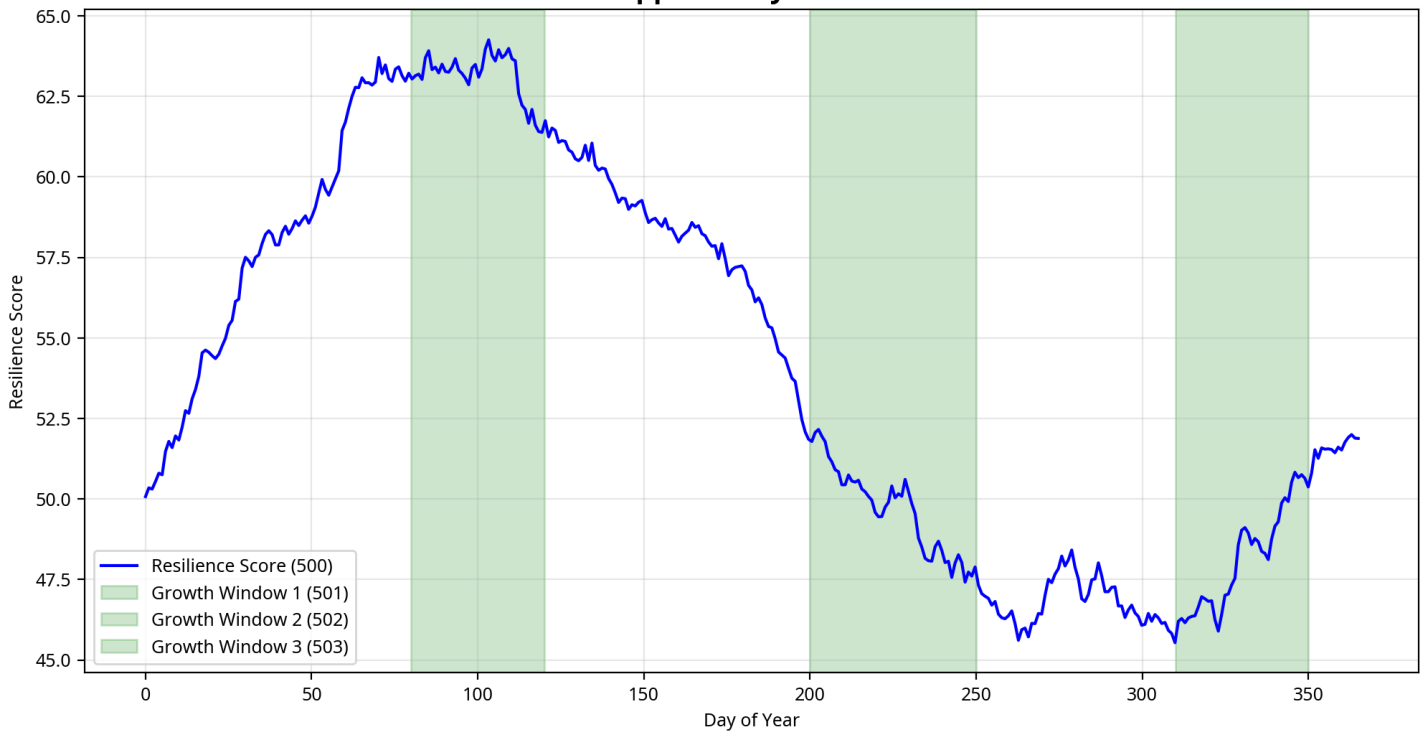


FIG. 6 - Hidden Markov Model State Transition Diagram

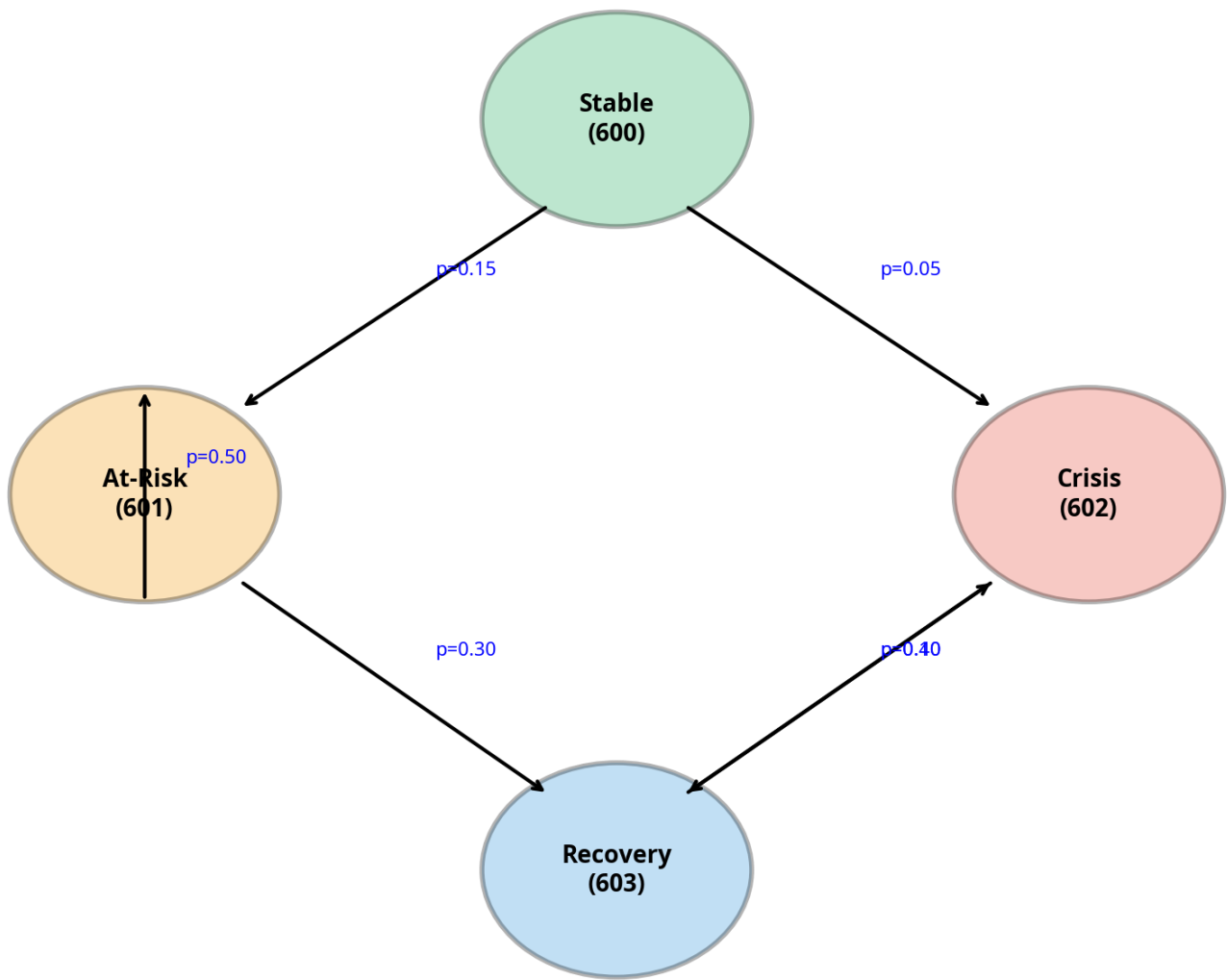


FIG. 7 - Wearable Biosensor Integration Pipeline

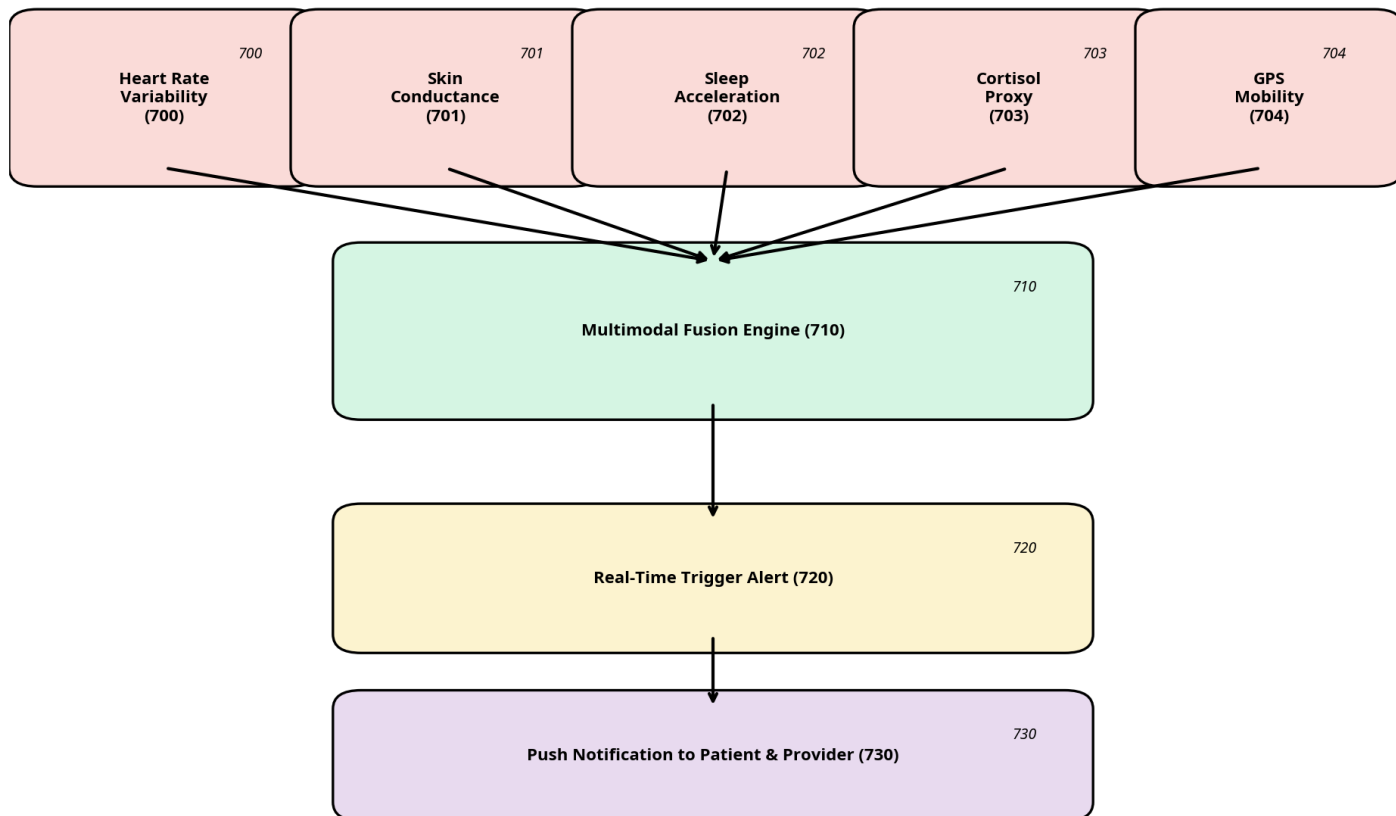


FIG. 8 - GPU-Accelerated Hawkes Process Computation Benchmark

